Kartik Choudhary

Computer Science Engineering Undergraduate

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SUMMARY

Computer Science Engineering student specializing in backend development with Java, Spring Boot, and MySQL. Experienced in building scalable AI-powered applications and optimization software. Proficient in object-oriented programming and data structures to deliver clean, efficient engineering solutions.

EXPERIENCE

SmartBridge & Google Cloud

Nov 2025 – Jan 2026

Google Cloud Generative AI Virtual Internship

- Completed intensive training in Generative AI, LLM APIs, and Google Cloud Platform development.
- Architected AI-powered backend systems using prompt engineering and cloud infrastructure.

EDUCATION

Lakshmi Narain College of Technology, Bhopal

Jan 2023 – Jan 2027

B.Tech, Computer Science Engineering

Bhopal · GPA: 8.76

St. Francis of Assisi School

Mar 2022 – Feb 2023

CBSE (Maths & Science), Maths & Science

GPA: 82.4%

SKILLS

Programming Languages: Java, SQL, JavaScript

Frontend: HTML5, CSS3, React, Jetpack Compose

Frameworks & Libraries: Spring Boot, Spring Security, REST APIs, JWT Authentication

Databases: MySQL, Database Design, JDBC

Tools & Software: Git, GitHub, Maven, Docker, Postman, JUnit, Mockito

Software Engineering Concepts:
RESTful API Design, Layered Architecture, Authentication & Authorization, SOLID Principles, Scalable Application Design

AI: AI-Powered Applications, LLM Fundamentals, Prompt Engineering

Professional Development: Generative AI, AI Agents, PowerShell

PROJECTS

MetroOpt AI – Intelligent Metro Fleet Allocation Platform

Java, Spring Boot, Google OR-Tools, JWT, Spring Security, MySQL, JUnit, Mockito, Docker

Developed a backend optimization platform to automate metro train fleet allocation and scheduling decisions.

- Developed a Java Spring Boot backend that optimizes allocation of 25+ metro trainsets using Google OR-Tools while satisfying maintenance, mileage, and operational constraints.
- Designed a modular REST API with JWT, Spring Security, and MySQL using SOLID principles and layered architecture.
- Developed optimization logic for train scheduling and maintenance constraints to drive large-scale allocation decisions.
- Created an AI-powered engine to translate complex optimization outputs into human-readable insights.

CineVerse - AI Powered Movie Discovery Platform

Spring Boot, REST APIs, Android, Web, JWT, Docker, JUnit, Mockito

Developed scalable backend supporting user authentication, watchlist management, and multi-client integration.

- Built scalable Spring Boot REST APIs for authentication, watchlist management, and multi-client integration.
- Architected modular service and repository layers using clean architecture and optimized database queries.
- Developed an AI recommendation engine processing movie metadata for personalized user suggestions.
- Implemented JWT security, containerized services with Docker, and authored JUnit and Mockito tests.

AWARDS

Solved 300+ Data Structures & Algorithms problems across coding platforms.

Selected for PwC Advisory Launchpad – Data & Tech Risk Program.